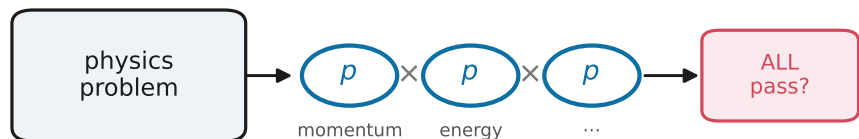


The constraint-penalty law: each conservation constraint multiplies the odds of a correct answer by ~ 0.69

a Constraint-satisfaction grading $\Rightarrow$ logistic in d_c



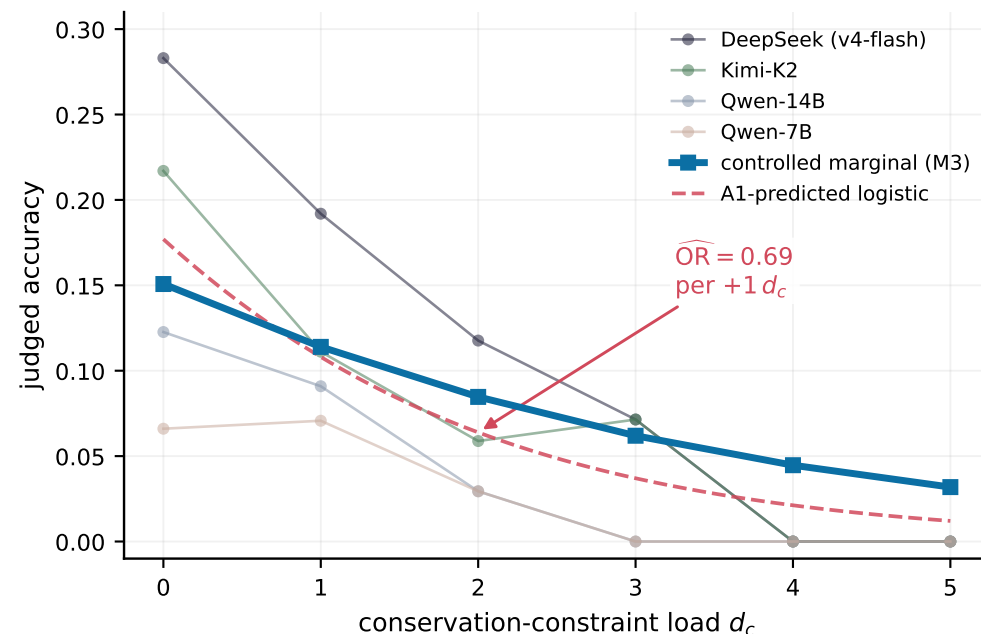
Correct only if every d_c constraint holds (A1):

$$P(\text{correct}) = \prod_{j=1}^{d_c} p_j = p^{d_c}$$

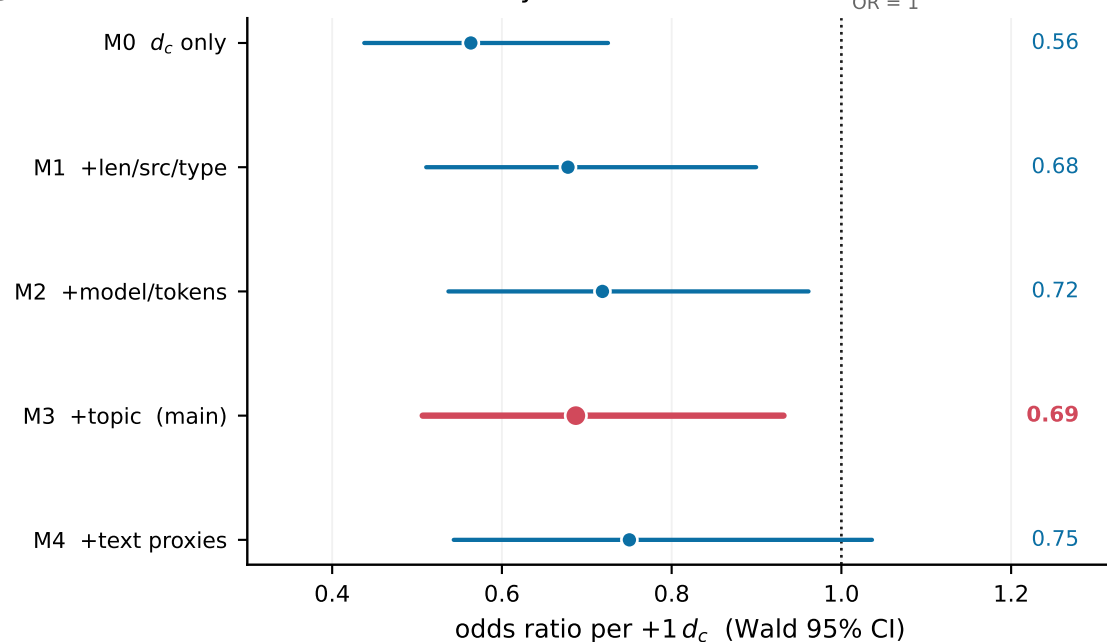
$$\text{logit } P = d_c \log \frac{p}{1-p} + \text{const}$$

$$\Rightarrow \text{OR} = e^\beta \approx 0.69 \approx 1/\sqrt{2} \text{ per constraint}$$

b Accuracy declines logarithmically with d_c



c Penalty is stable as controls accrue



M3 item-cluster bootstrap CI [0.44, 0.98]; proxy-expanded reaches 1.0 (see §4.4)

d Sign-stable across all 258 LIO folds

